



Network Security Defense Report

Submitted by: Team 4

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Destination	Application	Port	Protocol	DSCP	Traffic	Show Graph
172.18.22.26	www	80	TCP	Default	1.744 MB	
172.18.64.47	www	80	TCP	Default	778.114 KB	
203.149.210.179	www	80	TCP	Default	44.535 KB	

To monitor incoming web traffic and detect abnormal activity, we examined the firewall's application visibility logs. The dashboard recorded multiple HTTP connections on port 80 originating from both internal simulated networks (172.18.22.26 and 172.18.64.47) as well as an external host at **203.149.210.179**. All connections were correctly classified under the “**www**” application category using the TCP protocol, confirming that the firewall was successfully identifying and categorising web traffic. The traffic volume ranging from 44 KB to over 1.7 MB, also indicated that the visibility engine was actively capturing session sizes and maintaining real-time usage metrics. This level of monitoring is essential for detecting anomalies such as sudden spikes, repeated inbound attempts, or traffic patterns consistent with brute-force or DoS activity. By establishing this baseline visibility, the defence system is positioned to identify and respond to suspicious behaviour more effectively during an attack.

Top Conversation | Destination :: 192.168.70.10

Last hour | 2025-11-20 23:43 to 2025-11-21 00:43

Group By

None

Resolve IP

IN

OUT

Source	Application	Port	Protocol	DSCP	Traffic	Show Graph
172.18.64.47	cvd	8400	TCP	Default	0.000 Bytes	
172.18.64.47	ddi-tcp-1	8888	TCP	Default	0.000 Bytes	
172.18.64.47	ddrepl	4126	TCP	Default	0.000 Bytes	
172.18.64.47	domain	53	TCP	Default	0.000 Bytes	
172.18.64.47	eldim	31337	TCP	Default	0.000 Bytes	
172.18.64.47	emcrrmirccd	10004	TCP	Default	0.000 Bytes	
172.18.64.47	enpp	2968	TCP	Default	0.000 Bytes	
172.18.64.47	epmap	135	TCP	Default	0.000 Bytes	
172.18.64.47	finger	79	TCP	Default	0.000 Bytes	
172.18.64.47	fmsascon	16001	TCP	Default	0.000 Bytes	
172.18.64.47	fpo-fns	1066	TCP	Default	0.000 Bytes	
172.18.64.47	ftp	21	TCP	Default	0.000 Bytes	
172.18.64.47	ftp-data	20	TCP	Default	0.000 Bytes	

When reviewing the firewall's “Top Conversation” logs for the destination server **192.168.70.10**, we observed repeated connection attempts originating from **172.18.64.47**, an IP address that does not belong to our internal network range. This immediately flagged it as a **potential attacker or unauthorized external host**. The logs show this unknown IP probing a wide variety of ports and applications including domain (53), FTP (21/20), epmap (135), finger (79), and several high-numbered or uncommon ports (31337, 16001, 10004). Notably, all traffic entries show **0 bytes transferred**, which is characteristic of reconnaissance behaviour where an attacker is scanning and fingerprinting services without establishing full sessions. The presence of such diverse and unsolicited probes strongly suggests pre-attack enumeration activity, and the firewall's ability to capture and display these attempts provides valuable early-warning visibility for defensive monitoring and incident response.

Top Conversation Destination :: 172.18.64.20						
Last hour 2025-11-20 23:53 to 2025-11-21 00:53						
Source	Application	Port	Protocol	DSCP	Traffic	Show Graph
192.168.70.14	Unknown_App	*	Unknown	Default	39.523 MB	
203.149.210.179	Unknown_App	*	Unknown	Default	39.523 MB	
192.168.70.14	Unknown_App	*	Unknown	AF41	600.000 Bytes	
203.149.210.179	Unknown_App	*	Unknown	AF41	600.000 Bytes	

When analysing the “Top Conversation” logs for the destination **172.18.64.20**, we observed multiple traffic entries labelled as **Unknown_App**, with both the port and protocol fields marked as “Unknown.” Two of these connections originated from **203.149.210.179**, the same external IP previously flagged during earlier monitoring, while the remaining entries came from **192.168.70.14**, an internal host. The presence of unknown applications with unspecified ports suggests that these sessions did not match any predefined application signatures within the firewall, which is often indicative of **non-standard traffic, encrypted payloads, or probing activity**. Notably, the external IP generated approximately **39.5 MB** of traffic within the one-hour window, significantly higher than baseline activity. This volume, combined with the unresolved application type, reinforces the likelihood of potential scanning, tunnelling, or suspicious data exchange attempts. The firewall’s ability to surface this traffic even without full classification is critical for early anomaly detection and forms an important part of the network’s defensive visibility.

Top N Conversation

Last 30 minutes

Source	Destination	Application	DSCP	Traffic
101.100.179.144	192.168.90.10	https	Default	37.311 MB
172.18.64.47	203.149.210.177	www	Default	21.760 MB
203.149.210.179	172.18.64.20	Unknown_App	Default	12.010 MB
192.168.70.14	172.18.64.20	Unknown_App	Default	12.010 MB
101.100.179.144	203.149.210.180	https	Default	9.530 MB
172.18.64.47	192.168.70.10	www	Default	7.647 MB
172.18.64.20	203.149.210.179	Unknown_App	Default	6.062 MB
172.18.64.20	192.168.70.14	Unknown_App	Default	6.062 MB
192.168.101.2	192.168.90.10	palace-5	Default	5.627 MB
192.168.70.10	172.18.64.47	www	Default	3.519 MB

In the “Top N Conversation” view for the last 30 minutes, the firewall recorded several high-volume traffic flows that further reinforce earlier signs of suspicious behaviour on the network. Notably, the external IP **101.100.179.144** generated over **37 MB** of HTTPS traffic towards **192.168.90.10**, which is significantly higher than the network baseline and may indicate probing, automated requests, or attempted credential harvesting. The previously flagged suspicious host **172.18.64.47** also appears again, this time sending **21.7 MB** of HTTP (www) traffic to **203.149.210.177**, consistent with potential scanning or targeted enumeration. Meanwhile, the recurring unknown external IP **203.149.210.179** continues to generate **Unknown_App** traffic over **12 MB** towards **172.18.64.20**, which remains unclassified by the firewall’s application engine. The presence of multiple large, unclassified, or unexpected flows from both unknown external sources and non-authorised internal IPs suggests active reconnaissance or potential exploitation attempts within the monitored

interval. The firewall's ability to highlight these top conversations allows the defender to quickly identify abnormal patterns and prioritise further investigation.



Top Conversation Destination :: 192.168.16.17						
Last hour 2025-11-20 23:54 to 2025-11-21 00:54						
Source	Application	Port	Protocol	DSCP	Traffic	Show Graph
192.168.16.100	ICMP_App	*	ICMP	CS6	4.784 KB	

For the destination **192.168.16.17**, the firewall logs show a single notable traffic source: **192.168.16.100**, generating ICMP activity classified under **ICMP_App**. Although the traffic volume is relatively low at **4.784 KB**, the consistent appearance of ICMP packets especially when originating from only one host may indicate **network probing**, such as ping sweeps or host-discovery attempts. The DSCP value of **CS6**, which is typically associated with high-priority control traffic, further suggests that the packets were not part of normal user behaviour but could be automated or scripted. While ICMP itself is not harmful, repeated use of it during reconnaissance stages is common, and the firewall's ability to surface this activity provides defenders with early visibility into potential scanning or enumeration originating within the network.

Conclusion

Overall, the defence monitoring results show that our security controls were able to provide useful visibility into both normal and suspicious activities, allowing us to detect multiple indicators of attack behaviour early. From the firewall application and conversation logs, we established a baseline of legitimate web traffic, but also identified repeated reconnaissance patterns from unknown sources most notably 172.18.64.47 and 203.149.210.179 probing a wide range of ports with zero-byte sessions and generating unusually high volumes of unclassified Unknown_App traffic. The “Top N Conversation” view further confirmed abnormal spikes from external IPs (e.g., high HTTPS volume to internal monitoring hosts), reinforcing the likelihood of scanning or exploitation attempts rather than routine usage. Internally, ICMP activity toward 192.168.16.17 suggested host discovery behaviour consistent with early-stage reconnaissance.